Linac Access Control Interlock System
Test Report 2005-Nov-03

1.5.52.2 Rev. 0

Date: 2005-Dec-14

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REVISION HISTORY

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1.0 PURPOSE

This report outlines the results from testing of the Linac Access Control Interlock System (ACIS) conducted on November 3rd, 2005.

2.0 BACKGROUND

During a seven week shutdown between September 21st and November 3rd, 2005, the machine protection system for the Linac was upgraded. The ready chain for the Linac includes redundant ACIS interlocks. Both must be made up to permit operation of the components in question.

None of the changes to the machine protection system were expected to impact any part of the Linac ACIS before the point where the interface of the two systems occurs.

A verification and validation of the Linac ACIS had been completed recently [1]. In order to determine that

a. the lockup still performed as expected [2]

b. the gun and/or Linac RF could not be enabled even if zones 1-5 are locked up unless BST0003-01 is in line and a beam target in the Linac is chosen, and

c. any unlocked zones, from 1-5, prevent the operation of both the gun triggers and Linac RF

A test procedure was generated and performed by the HSE department prior to the start of normal operations (see Appendix A).

3.0 VERIFICATION AND VALIDATION RESULTS

3.1 SUMMARY

3.1.1 ACIS Zone Lockup Requirements

Each of zones 1-5 was locked up successfully using the Linac/LTB Lockup Procedure [2]. During the test, the lockup for each zone was breached, one zone at a time, by opening an interior gate, opening the entrance gate or door, or pressing an emergency off switch (EOS). After a zone lockup was breached, the zone was relocked prior to proceeding with the next test.

All 5 zones behaved as expected in that the lockup was breached by each of the methods chosen. While pressing an EOS in zone 2, it was observed that the lockup for zone 1 dropped out in addition to zone 2. While pressing an EOS in zone 4, the lockup for zone 5 dropped out in addition to zone 4.

The lockup light in zone 2 labeled LH6, located near the bottom of the zone 2 stairwell, was found non-functioning. The installation of the new machine protection terminal block panel in zone 2 has resulted in a condition where the zone lockup light LH2 is now obscured by the door of the panel if the door is left open.

3.1.2 ACIS Linac Ready Chain Requirements

This test required that all zones be locked and the keys turned in the control room Kirk Key Panel. The first test required showing that with all ACIS interlocks made up and the inline beam stop (BST0003-01) out (not in the beam path), the electron gun and the Linac RF could not be enabled. In all cases the gun functionality was checked by attempting to enable the gun [3], and the RF was checked by observing the voltage at the modulator.

The result of this test showed that the gun and RF interlocks were not inhibited. Testing was terminated and the problem was reported to the CID department. They investigated and found that the interface tie-in between the machine protection system and the ACIS had not been completed correctly. The problem was fixed, and when testing was restarted it was found that the gun and RF ACIS interlocks were inhibited when the beam stop was out. The beam stop was then put inline and the control room ACIS keys turned to the disable position. The gun and Linac RF were again inhibited by the ACIS interlocks.

For the second part of the test, each of zones 1-5 was locked up. Each zone entrance gate or door was opened in sequence to confirm that the ACIS interlocks prevented operation of the electron gun and the Linac RF. Each zone was locked up and all interlocks reset prior to testing the next zone.

4.0 RECOMMENDATIONS

1) Although the effect of pressing an EOS did not create an unsafe condition, the fact that pressing an EOS in one zone could cause the lockup to drop in another zone should be confirmed as correct behaviour of the system (MKS 766).
2) Lockup light LH6 needs to be replaced as soon as possible (MKS 767).
3) A notice should be put on the inside and outside of the new machine protection terminal block panel (P0009-07) informing personnel that the panel door must be closed prior to zone lockup.

5.0 REFERENCE MATERIALS

[1] Linac Access Control Interlock System Validation and Verification Report 1.5.52.1.Rev.0-Cubbon

[2] Linac to LTB1 Lockup Procedure 2.7.37.2.Rev.1B -West

[3] Linac Setup Manual 1.9.36.1, Rev. 2 – Shen, Silzer

6.0 ATTACHMENTS

[1] Linac ACIS Testing Requirements Checklist

7.0 APPENDIX

7.1 LINAC ACIS TESTING REQUIREMENTS

Recent changes to the machine protection of the Linac have impacted the enable requirements for the gun and the Linac RF. The signal from the Linac hardwired and Linac PLC systems, along with the gun emergency off, are required together with the machine protection chain in order for the gun or the Linac RF to be enabled.

In order to confirm that the Linac ACIS has not been affected by the recent work on the machine protection system, the following tests must be performed:

1) ACIS Zone lockup requirements

To confirm hardwire and PLC sides of the Linac ACIS, each zone should be locked up and the lockup breached by:

a. opening an interior gate to the zone
b. opening the main entrance gate to the zone
c. pressing an emergency off button in each zone (disconnect control room horn once it has been shown to be operational – reconnect horn and confirm operation when testing complete).

TEST 1 ACIS	**Zone 1**	**Zone 2**	**Zone 3**	**Zone 4**	**Zone 5**
Interior Gate					
EOS					
Entrance Gate					

2) ACIS Linac Ready Chain Requirements

To confirm that a breach of the lockup of any of zones 1-5 will defeat the ready chain required for enabling the Gun and enabling the RF, the following steps are required:

a. Lockup zones 1-5 and enable keys in control room (BST 0003-01 OUT), verify that the Gun and RF are disabled*.
b. Lockup zones 1-5 and enable keys in control room (BST 0003-01 IN)
c. Verirfy the Gun and Linac RF operation is permitted.
d. Breach lockup by turning keys off in control room
e. Verify the Gun and Linac RF are disabled.
f. Re-enable keys in control
g. Verify Gun and Linac RF operation is permitted
h. Sequentially breach lockup of zones 1-5 using spare ACIS keys, verifying that the Gun and Linac RF are disabled when the lockup is breached for each zone. Lockup each zone and verify that the RF and Gun operation is permitted before the next test.

* AOD assistance may be required.

Test 2 Ready Chain	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
BST0003-01 IN	**Zone 1 Gun/RF**	**Zone 2 Gun/RF**	**Zone 3 Gun/Rf**	**Zone 4 Gun/RF**	**Zone 5 Gun/RF**
Open Entrance Gate					
Close gate, Reset Linac/Gun					

<u>Control Room Keys</u>					

LINAC ACIS Testing Requirements

Recent changes to the machine protection of the LINAC have impacted the enable requirements for the gun and the LINAC RF. The signal from the LINAC Hardwire and LINAC PLC, along with the gun emergency off, are required together with the machine protection chain in order for the gun or the LINAC RF to be enabled.

In order to confirm that the LINAC ACIS has not been affected by the recent work on the machine protection system, the following tests must be performed:

1) <u>ACIS Zone lockup requirements</u>

To confirm hardwire and PLC sides of the LINAC ACIS, each zone should be locked up and the lockup breached by:

a. opening an interior gate to the zone
b. opening the main entrance gate to the zone
c. pressing 1 emergency off button in each zone (disconnect control room horn once it has been shown to be operational).

LH6 burnt out (outside bottom of stairwell door)
LH2 hidden by new Panel door

TEST 1 ACIS	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Interior Gate	✓	✓	✓	✓	✓
EOS	✓	✓	✓	✓	✓
Entrance Gate	✓	✓	✓	✓	✓

Zone 1 lockup mystery drop
Zone 5 lockup " "

<u>2) ACIS LINAC Ready Chain Requirements</u>

To confirm that a breach of the lockup of any of zones 1-5 will defeat the ready chain required for enabling the Gun and enabling the RF, the following steps are required:

d. Lockup zones 1-5 and enable keys in control room (BST 0003-01 OUT), verify that the Gun and RF cannot be enabled*. ✓
e. Lockup zones 1-5 and enable keys in control room (BST 0003-01 IN)
f. Verirfy the Gun and LINAC RF can be enabled ✓
g. Breach lockup by turning keys off in control room ✓
h. Verify the Gun and LINAC RF cannot be enabled ✓
i. R-enable keys in control ✓
j. Verify Gun and LINAC RF can be enabled ✓
k. Sequentially breach lockup of zones 1-5 using spare ACIS keys, ✓ verifying that the Gun and LINAC RF cannot be enabled when the lockup is breached for each zone. Lockup each zone and verify that the RF and Gun can be enabled before the next test.

* AOD assistance may be required.

Test 2 Ready Chain	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
BST 0003-01 IN	**Zone 1 Gun/RF**	**Zone 2 Gun/RF**	**Zone 3 Gun/Rf**	**Zone 4 Gun/RF**	**Zone 5 Gun/RF**
Open Entrance Gate	✓ ✓	✓ ✓	✓ ✓	✓ ✓	✓ ✓
Reset Linac/Gun	✓ ✓	✓ ✓	✓ ✓	✓ ✓	✓ ✓

Control Room Keys	Gun off	R F off			
BST 0003-01 IN	✓	✓			
OUT	✓	✓			

[signature] Nov 3/05

Grant Cubbon

[signature] Nov 8/05

Allen Hodges

~~These forms do not apply to the Linac.~~

6.0 ATTACHMENTS/FORMS

6.1 Non-Conformance Report

Date: Nov 7/05 CLS Problem ID #: 766

Device or I/O Point: LINAC Lockup

Failure Description: ① Pressing an EOS in zone 2 dropped lockup of zone 1 ② Pressing EOS in 4 dropped lockup in zone 5	Initials
	SC
Cause: The older linac lockup system does not by design behave the same way as the newer BR1/SR1/BL system.	EM.
Resolution: No safety implication as a result of the non-conformance. Therefore no change until next planned upgrade	EM/MB

closed

Cc: HSE Manager
CLS Files

6.0 ATTACHMENTS/FORMS

6.1 Non-Conformance Report

Date: Nov 7/05   CLS Problem ID #: 768

Device or I/O Point: LINAC ACIS LH2

Failure Description:	Initials
Lockup light LH2 in zone 2 is hidden by machine protection TB panel door when the door is left open.	SC

Cause: N/A

Resolution:	Initials
~~Lockup procedure upd modified to include a step ensuring panel is closed as part of Zone 2 lockup.~~	~~[illegible]~~
Circulate a memo advising all lockup-trained personnel to ~~keep door closed. A~~ ensure panel door is closed. ~~That~~ Further action is deferred pending HSE evaluation. ~~A~~ Signage will be posted to the same effect on panel door.	MB

Cc: HSE Manager
CLS Files

Deferred